
DSAINT
DISTRIBUTED SOVEREIGN AI
INFRASTRUCTURE NODES TECHNOLOGY
CONCEPT
DEVELOPMENT
DOCUMENT
Building Germany's
Sovereign AI Infrastructure
Backbone
DIGITAL SOVEREIGNTY
Infrastructure owned
and controlled in Germany
DISTRIBUTED BY DESIGN
Resilient, scalable and
regionally interconnected
SUSTAINABLE FUTURE
Energy-efficient operations
for a responsible tomorrow
COMMUNITY POWERED
Citizens, institutions and industry
building the future together
CONTACT
Ernest Repko
Founder – DSAINT
info@dsaint.eu

DSAINT

DISTRIBUTED SOVEREIGN AI INFRASTRUCTURE NODES TECHNOLOGY

CONCEPT DEVELOPMENT DOCUMENT

Updated Strategic & Technical Edition

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From a national infrastructure vision to a focused, team-based technology initiative: AI workload orchestration, trusted federation and eventual physical validation.

Prepared by Ernest Repko
Founder – DSAINT
info@dsaint.eu | www.dsaint.eu

Updated on the basis of the original DSAINT Concept Development Document (Version 2.0) and subsequent strategic feedback.

Document Basis and Update Principles

The uploaded source document is the baseline for this edition. Its core terminology, architecture and long-term vision have been retained. The update introduces a narrower near-term development strategy following the subsequent BMDS discussion described by the project founder.

The original document presents DSAINT as a sovereign AI infrastructure initiative combining Core Infrastructure Nodes, Research & Innovation Nodes, Enterprise Nodes and Citizen AI Nodes, with Reverse Co-Hosting and a Community Compute Federation at its center. It specifies a planned German Core Infrastructure Network of twelve locations and Demonstrator Node 01 in Bedburg-Hau. These elements remain part of the long-term concept.

The principal update is strategic: DSAINT should not be presented as something one founder can build alone. The immediate objective is to develop a technically demonstrable orchestration and federation prototype and integrate the project with experienced teams already active in Germany and Europe.

Source-derived content is retained as the DSAINT baseline; updated sections explicitly distinguish immediate technical priorities from long-term concepts.

1. Executive Summary

DSAINT (Distributed Sovereign AI Infrastructure Nodes Technology) is a concept and technology framework for distributed sovereign AI infrastructure in Germany and, ultimately, Europe. The original concept combines professional Core Infrastructure Nodes, Research & Innovation Nodes, Enterprise Nodes and Citizen AI Nodes into a unified federation.

The updated strategy keeps this long-term architecture but changes the development sequence. The immediate technical focus becomes AI workload orchestration across heterogeneous, geographically distributed and independently operated compute resources. A small software-defined federation prototype becomes the first major proof point. Demonstrator Node 01 remains the planned physical validation environment rather than the first required proof of feasibility.

This approach responds directly to the practical reality that Germany already has organizations working on related distributed AI infrastructure problems. DSAINT therefore seeks to contribute through technical depth, measurable experimentation, collaboration and integration into an existing ecosystem.

Long-term ambition: distributed sovereign AI infrastructure. Near-term proof: a working DSAINT federation and workload-orchestration prototype.

2. Vision and Mission

2.1 Vision

To contribute to a resilient, distributed and sovereign AI infrastructure ecosystem in Germany by developing technologies and operating concepts that allow heterogeneous AI compute resources to be securely orchestrated across organizational and geographic boundaries.

2.2 Mission

To research, prototype and validate the technical principles required to federate trusted AI compute resources through intelligent workload orchestration, while establishing a credible path toward physical sovereign AI infrastructure.

2.3 Long-Term DSAINT Principle

The original DSAINT slogan remains a guiding concept:

“Reverse co-hosting is the future of Germany’s independent AI – a future in which everyone can participate.”

In this updated edition, Reverse Co-Hosting is treated as a long-term research and participation model whose technical, economic, legal and security assumptions must be validated.

3. The Challenge

The source document identifies AI as increasingly critical infrastructure and highlights dependence on foreign AI infrastructure, growing compute demand, rising costs of advanced AI workloads, limited public participation and the need for secure sovereign AI ecosystems.

- Concentration of advanced compute capacity can create strategic dependency.
- AI workloads require increasingly specialized accelerators, networking, storage, power and cooling.
- Research, public-sector and industrial users need reliable access to sovereign or trusted compute resources.
- Distributed resources are difficult to coordinate when hardware, operators, policies and locations differ.
- A future community participation model requires strong security, trust, accounting and governance mechanisms.

3.1 Updated Problem Definition

How can AI workloads be intelligently, securely and efficiently orchestrated across heterogeneous, geographically distributed and independently operated compute nodes?

4. The DSAINT Solution

The DSAINT solution remains a distributed infrastructure federation, but its development is organized around an enabling technology stack rather than around immediate construction of a national network.

Component	Long-term role	Near-term status
Core Infrastructure Network	Professional sovereign AI backbone	Blueprint / future physical validation
Research & Innovation Nodes	Universities, HPC and research participation	Target partner environment
Industrial & Enterprise Nodes	Industrial, healthcare, utility and enterprise compute	Target partner environment
Community Compute Federation	Community-operated resources through Reverse Co-Hosting	Research concept / controlled validation
DSAINT Orchestrator	Federation brain: discovery, policy, scheduling and monitoring	Primary prototype focus

5. Updated Four-Layer Architecture

The four-layer architecture from the original document is retained, with an explicit orchestration and federation control plane added across the layers.

Layer	Function
Layer 1 – Core Infrastructure Network	Environmentally responsible Tier III-oriented AI infrastructure nodes providing sovereign backbone services.
Layer 2 – Research & Innovation Nodes	Universities, HPC facilities and research institutions contributing compute, expertise and validation.
Layer 3 – Industrial & Enterprise Nodes	Industry, healthcare, utilities and enterprises providing or consuming trusted AI compute.
Layer 4 – Community Compute Federation	Citizen AI Nodes participating through the researched

	Reverse Co-Hosting model.
Cross-layer DSAINT Orchestration & Federation Plane	Node identity, capability discovery, workload submission, scheduling, policy, security, observability and contribution accounting.

AI workload → requirements & policy → resource discovery → candidate nodes → scheduling → secure execution → monitoring → results → accounting

6. AI Workload Orchestration – Immediate Technical Core

The project's immediate technical specialization is AI workload orchestration. This creates a tractable engineering problem that can be researched and demonstrated before large physical infrastructure is deployed.

Research area	Initial question
Workload scheduling	Which node should execute a workload under current constraints?
GPU allocation	How should accelerator capacity be allocated efficiently?
Heterogeneity	How should different GPU generations and node capabilities be handled?
Data locality	How should network, storage and latency influence placement?
Security policy	Which workloads may execute on which trust class of node?
Reliability	What happens when a node becomes unavailable?
Energy awareness	Can workload placement account for energy efficiency?
Observability	How can capacity, health and performance be measured across the federation?
Operator control	How can each node operator retain control over permitted workloads?

7. DSAINT Federation Prototype 01

The first major technical artifact should be a small software-defined federation that can run on a laptop, workstation, virtual machines and/or a small number of physical compute nodes.

- Node registration and identity
- Node capability discovery
- CPU/GPU resource inventory
- Workload submission interface
- Queue and scheduling engine
- Policy-based workload placement
- Node health monitoring
- Execution status and logging
- Failure detection and rescheduling
- Basic contribution accounting

- Performance dashboard

7.1 Prototype Progression

1. Simulate multiple nodes on one development machine.
2. Represent nodes using containers or virtual machines.
3. Connect two or more independent compute environments.
4. Introduce GPU-aware scheduling.
5. Add workload and node policy classes.
6. Introduce observability and failure handling.
7. Test controlled federation between independently operated nodes.
8. Benchmark selected AI workloads and publish measurable results.

8. Community Compute Federation and Reverse Co-Hosting

The source document describes a Community Compute Federation inspired by the anthill principle, where many small contributions can form a collective resource. It proposes participation by citizens and organizations and compensation based on measured contribution.

The updated DSAINT position is deliberately more rigorous: the model is a research hypothesis until security, workload isolation, economics, legal responsibility, energy accounting, data protection and reliability have been validated.

Question	Required validation
Trust	How is a node authenticated and assigned a trust level?
Isolation	How are workloads isolated from the node operator and other workloads?
Data	Which workloads and data classes are permitted?
Security	How is hardware/software integrity established?
Reliability	How is node churn handled?
Economics	How is contribution measured and compensated?
Energy	How are additional energy costs measured?
Liability	Who is responsible for workload and infrastructure incidents?

Reverse Co-Hosting remains strategically important, but DSAINT will earn credibility by validating it incrementally rather than assuming it is already solved.

9. Demonstrator Node 01

The source document specifies Demonstrator Node 01 in Bedburg-Hau, Germany, as the operational blueprint for future DSAINT deployments.

Parameter	Source baseline
Location	Bedburg-Hau, Germany
Initial capacity	2 MW
Expansion capacity	6 MW
Infrastructure standard	Tier III

Target PUE	≤ 1.20
Cooling	Direct-to-Chip Liquid Cooling
Redundancy	N+1
Estimated CAPEX	€43 million

Updated role: Node 01 is a future physical validation platform. DSAINT does not need to construct Node 01 before its orchestration principles can be tested. Software and federation experiments should establish technical evidence first.

- Validate high-density AI infrastructure concepts
- Integrate the DSAINT orchestration platform
- Measure power, cooling and compute performance
- Test secure connectivity and federation
- Provide a reference architecture for future deployments

10. German Core Infrastructure Network

The original DSAINT concept proposes twelve strategically distributed Core Infrastructure Nodes. The locations remain a planning hypothesis and should be validated against power, fiber connectivity, land, cooling potential, demand, security, permitting and economic feasibility.

No.	Planned location
01	Bedburg-Hau
02	Lindau
03	Hamelin
04	Schwäbisch Hall
05	Eberswalde
06	Wismar
07	Limburg an der Lahn
08	Wernigerode
09	Bad Kreuznach
10	Bautzen
11	Gotha
12	Husum

The network is intended to provide geographic resilience, lower-latency regional access and distributed sovereign capacity. Detailed site selection should follow a future feasibility study.

11. Security and Compliance

The original document adopts security-by-design principles and identifies BSI IT-Grundschutz, dedicated secure networks, encrypted standard protocols, strict identity and access management, GDPR compliance and state/federal regulatory compliance.

- Federated identity and node authentication
- Role- and policy-based access control
- Encrypted communications
- Workload isolation

- Audit logging and observability
- Security classification of nodes and workloads
- Protection of sensitive data and credentials
- Security testing before broader federation

The technical prototype should implement only the security mechanisms that can be credibly demonstrated at its maturity level, while maintaining a clear path toward production-grade compliance.

12. Sustainability

The source concept specifies a target PUE of ≤ 1.20, heat recovery, direct-to-chip liquid cooling, municipal heat integration, future renewable-energy integration and sustainable infrastructure design.

- Measure rather than assume energy efficiency.
- Investigate workload placement based on energy and thermal conditions.
- Use direct-to-chip liquid cooling for appropriate high-density deployments.
- Evaluate waste-heat utilization where local conditions support it.
- Consider renewable-energy integration in future physical nodes.

13. Governance

The original governance model consists of a Supervisory Board, Executive Team, Technical Governance Board, Security Committee and Sustainability Committee.

Governance element	Updated purpose
Supervisory Board	Long-term strategic oversight when organizational maturity requires it.
Executive Team	Operational and partnership leadership.
Technical Governance Board	Architecture, standards, engineering quality and research direction.
Security Committee	Security, trust, compliance and risk.
Sustainability Committee	Energy, cooling, heat recovery and environmental objectives.

At the current stage, these bodies should be regarded as the target governance structure rather than implying that all bodies are already formally constituted.

14. Ecosystem Integration Strategy

A central update is the explicit recognition that DSAINT operates in a mature ecosystem. Germany already contains companies and research teams working on distributed AI infrastructure, orchestration, HPC, sovereign cloud and related technologies.

DSAINT will therefore pursue collaboration and technical integration rather than presenting itself as an isolated replacement for existing initiatives.

- Map existing German and European solutions.
- Identify technical gaps and complementary capabilities.
- Approach experienced teams as potential collaborators and employers.

- Use the prototype to demonstrate concrete technical contribution.
- Seek research and industrial validation before large-scale infrastructure commitments.

The strategic objective is not to build DSAINT alone; it is to make DSAINT technically useful enough to become part of the ecosystem building sovereign AI infrastructure.

15. DSAINT Differentiation

DSAINT should not rely on the claim that distributed AI infrastructure itself is novel. Its differentiation should be established through research and evidence.

Potential DSAINT combination	Question to validate
Physical sovereign nodes + federation	Does combining owned infrastructure with federation create a meaningful operational advantage?
AI workload orchestration + sovereignty	Can scheduling enforce sovereignty and trust requirements?
Professional + research + enterprise + community layers	Can heterogeneous operators participate under a common technical model?
Reverse Co-Hosting	Can distributed community resources safely contribute useful AI capacity?
Energy-aware federation	Can placement improve energy and infrastructure efficiency?

A formal competitive and research landscape study should be maintained as a living DSAINT workstream.

16. Ownership and Business Model Direction

The long-term DSAINT ownership concept separates infrastructure ownership from compute hardware ownership and from software/federation governance. This allows municipalities, infrastructure owners, operators, technology partners and participating compute providers to have distinct roles.

At the present stage, the most important objective is technical validation. Legal entities, investment structures and detailed compensation mechanisms should be refined only as the technical and partner ecosystem develops.

Area	Near-term approach	Long-term direction
Infrastructure	Partner/site feasibility	Distributed node ownership
Software	Prototype and research	DSAINT orchestration platform
Compute	Test heterogeneous resources	Federated capacity
Community	Controlled research	Reverse Co-Hosting
Revenue	Validate use cases	Infrastructure/services/federation models
Governance	Project-level coordination	Formal multi-stakeholder governance

17. Updated Development Roadmap

Phase	Period	Primary objective	Evidence

1. Technical foundation	2026–2027	AI infrastructure skills and laboratory	Working local test environment
2. Federation Prototype 01	2027	Node discovery, scheduling, execution, monitoring	Multi-node prototype
3. Technical validation	2027–2028	GPU, heterogeneity, policies and failure testing	Benchmarks and technical report
4. Ecosystem integration	2027–2029	Join/collaborate with experienced teams	Partner/team participation
5. Demonstrator preparation	2028–2030	Translate validated architecture to physical environment	Node integration design
6. Node 01 validation	Target 2027+	Physical infrastructure demonstrator	Operational demonstrator
7. Core network expansion	Long term	Regional distributed infrastructure	Validated deployments
8. European expansion	Long term	Cross-border federation	European ecosystem

The original source roadmap proposed 2027 Node 01, 2028 Citizen Node Program, 2030 three-node expansion, 2032 six-node ecosystem, 2034 twelve operational Core Nodes and 2036 European expansion. This edition retains those milestones as an aspirational long-term scenario but introduces technical-gate milestones before them.

18. Success Criteria

- A functioning DSAINT multi-node prototype exists.
- AI workloads can be submitted, scheduled and monitored across multiple nodes.
- At least one heterogeneous-resource scheduling experiment produces measurable results.
- Security and trust policies are demonstrated at prototype level.
- The project has completed a documented German/European landscape and gap analysis.
- DSAINT has entered a practical collaboration with an experienced technical team or organization.
- Demonstrator Node 01 has a technically grounded integration path.
- The long-term federation model is supported by evidence rather than assumptions.

19. Key Risks and Mitigations

Risk	Mitigation
Scope too broad	Make workload orchestration and federation the near-term core.
Existing solutions are more mature	Map the landscape and seek complementary positioning.
Founder lacks large-scale execution experience	Join or collaborate with experienced teams.
Community compute is technically difficult	Validate in controlled stages before public deployment.
Large infrastructure requires substantial capital	Separate software validation from physical deployment.
Security/privacy barriers	Use trust classes, isolation and policy controls.
Overstatement of novelty	Use evidence-based differentiation and competitive analysis.
Insufficient technical evidence	Build, measure and document a working prototype.

20. Team and Founder Development

The project now explicitly assumes multidisciplinary teamwork. The founder's development path is aligned with the technical core.

- Linux and systems administration
- Networking
- Python
- Git/GitHub
- Docker
- Kubernetes
- GPU computing / CUDA fundamentals
- GPU scheduling
- Slurm / HPC
- Distributed systems
- Prometheus / Grafana
- AI workload orchestration
- Security fundamentals

The intended outcome is not to become a sole expert in every discipline, but to become a technically credible contributor and future coordinator of a multidisciplinary sovereign AI infrastructure project.

21. DSAINT Core Documentation Set

Document	Purpose
DSAINT Concept Development Document	Vision, architecture, strategy and long-term development framework.
DSAINT Technical Research Roadmap	Research questions, learning plan, experiments and validation.
DSAINT Federation Prototype Specification	Concrete software architecture and implementation requirements.
Demonstrator Node 01 Blueprint	Physical infrastructure reference and future validation environment.
Technical Architecture	Detailed system architecture across physical, software and federation layers.
Business Model	Long-term operating, ownership and value-creation structure.
Financial Model	Capital, operating and scenario assumptions for future deployments.

These documents should evolve from evidence generated by the prototype and partner work rather than from assumptions alone.

22. Immediate Next Actions

9. Complete a structured German and European competitive/research landscape.
10. Finalize the AI workload orchestration learning program.
11. Set up the DSAINT Federation Prototype 01 laboratory.
12. Define 3–5 measurable experiments.

13. Implement node discovery and a first scheduling policy.
14. Add monitoring and failure handling.
15. Identify technical teams working on adjacent problems.
16. Seek practical collaboration or employment within an experienced infrastructure team.
17. Use prototype evidence to refine the SPRIND proposal.
18. Keep Node 01 as the long-term physical demonstrator.

23. Conclusion

DSAINT remains an ambitious long-term concept for a distributed sovereign AI infrastructure ecosystem connecting professional infrastructure, research, industry and community resources.

The updated development strategy makes the project more executable: the first objective is not to construct a national network, but to develop and validate the technical mechanisms that could make such a network possible.

By focusing on AI workload orchestration, trusted federation, measurable experiments and integration with experienced German and European teams, DSAINT can progress from a broad infrastructure vision toward a credible technical contribution.

From vision → to prototype → to team integration → to validated infrastructure → to sovereign AI federation.

24. Contact

Item	Information
Project	DSAINT – Distributed Sovereign AI Infrastructure Nodes Technology
Founder	Ernest Repko
Location	Bedburg-Hau, Germany
Email	info@dsaint.eu
Website	www.dsaint.eu
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